

## Initial Project Planning Template

|               |               |
|---------------|---------------|
| Date          | 5 July 2026   |
| Team ID       |               |
| Project Name  | Rising Waters |
| Maximum Marks | 5 Marks       |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                          | Story Points | Priority | Team Members | Sprint Start Date | Sprint End Date (Planned) |
|----------|-------------------------------|-------------------|------------------------------------------------------------|--------------|----------|--------------|-------------------|---------------------------|
| Sprint-1 | Data Collection               | USN-1             | Download the flood prediction dataset.                     | 2            | High     | Khushwanth   | 24 June 2026      | 24 June 2026              |
| Sprint-1 | Visualizing & Analysing Data  | USN-2             | Import the required Python libraries.                      | 1            | High     | Khushwanth   | 24 June 2026      | 24 June 2026              |
| Sprint-1 | Visualizing & Analysing Data  | USN-3             | Read and explore the dataset                               | 2            | High     | Khushwanth   | 24 June 2026      | 25 June 2026              |
| Sprint-1 | Visualizing & Analysing Data  | USN-4             | Perform univariate, multivariate and descriptive analysis. | 3            | High     | Khushwanth   | 25 June 2026      | 26 June 2026              |
| Sprint-2 | Data Pre processing           | USN-5             | Handle missing values and outliers (IQR Capping).          | 3            | High     | Khushwanth   | 26 June 2026      | 27 June 2026              |
| Sprint-2 | Data Pre                      | USN-6             | Handle categorical values and                              | 2            | High     | Khushwanth   | 27 June 2026      | 28 June 2026              |

|          |                      |        |                                                                         |   |        |            |              |              |
|----------|----------------------|--------|-------------------------------------------------------------------------|---|--------|------------|--------------|--------------|
|          | processing           |        | perform feature scaling.                                                |   |        |            |              |              |
| Sprint-2 | Data Pre processing  | USN-7  | Split the dataset into training and testing sets.                       | 1 | High   | Khushwanth | 28 June 2026 | 28 June 2026 |
| Sprint-3 | Model Building       | USN-8  | Build the Decision Tree model.                                          | 2 | High   | Khushwanth | 29 June 2026 | 29 June 2026 |
| Sprint-3 | Model Building       | USN-9  | Build the Random Forest model.                                          | 2 | High   | Khushwanth | 29 June 2026 | 30 June 2026 |
| Sprint-3 | Model Building       | USN-10 | Build the KNN model.                                                    | 2 | Medium | Khushwanth | 30 June 2026 | 30 June 2026 |
| Sprint-3 | Model Building       | USN-11 | Build the XGBoost model.                                                | 3 | High   | Khushwanth | 1 July 2026  | 1 July 2026  |
| Sprint-4 | Model Evaluation     | USN-12 | Compare all models and select the best-performing model (XGBoost).      | 2 | High   | Khushwanth | 2 July 2026  | 2 July 2026  |
| Sprint-4 | Model Evaluation     | USN-13 | Evaluate the final model and save it as a Joblib (.save) file.          | 1 | High   | Khushwanth | 2 July 2026  | 2 July 2026  |
| Sprint-4 | Application Building | USN-14 | Build the HTML pages for the web application.                           | 3 | High   | Khushwanth | 3 July 2026  | 3 July 2026  |
| Sprint-4 | Application Building | USN-15 | Develop the Flask application (app.py) and integrate the trained model. | 4 | High   | Khushwanth | 3 July 2026  | 4 July 2026  |
| Sprint-4 | Application Building | USN-16 | Run, test and validate the flood prediction application.                | 2 | High   | Khushwanth | 4 July 2026  | 4 July 2026  |